NLP Question Paper — Detailed Step-by-Step Solutions (Q1–Q9)

Q1. CYK Parsing (a, b, c) — Step-by-step

For each sub-question we apply the CYK algorithm. CYK requires the grammar in Chomsky Normal Form (CNF): productions must be A → BC or A → a. We show the table filling (span by span) and conclude whether S appears in the top cell.

Q1(a) Given grammar and sentence: "the dog chased a cat"

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(i) CNF suitability:
- Each production must be either binary nonterminals A → B C or terminal A → a.
- The given grammar is suitable except NP → N is a unit production; we handle it by
unit-closure: whenever N appears we also add NP.

(ii) CYK table construction for the sentence with tokens
1=the,2=dog,3=chased,4=a,5=cat.
We build triangular table with spans length 1..5.

Span 1 (terminals):
(1,1) 'the' -> Det
(2,2) 'dog' -> N (and by NP → N add NP)
(3,3) 'chased' -> V
(4,4) 'a' -> Det
(5,5) 'cat' -> N (and NP)

So:
(1,1): {Det}
(2,2): {N, NP}
(3,3): {V}
(4,4): {Det}
(5,5): {N, NP}

Span 2 (length=2):
(1,2) 'the dog' : Det + N -> NP (rule NP → Det N) => {NP}
(2,3) 'dog chased' : {N,NP} + {V} -> no matching A→B C => {}
(3,4) 'chased a' : {V} + {Det} -> no match => {}
(4,5) 'a cat' : Det + N -> NP => {NP}

Span 3:
(1,3) 'the dog chased' : splits (1,1)+(2,3) and (1,2)+(3,3).
- (1,2)+(3,3): NP + V -> no rule => {}
(2,4) 'dog chased a' : no valid splits -> {}
(3,5) 'chased a cat' : V + NP -> VP (rule VP → V NP) => {VP}

Span 4:
(1,4) 'the dog chased a' : no valid combining results => {}
(2,5) 'dog chased a cat' : (2,2)+(3,5) => NP + VP yields S (S → NP VP) => {S}

Span 5 (whole sentence):
(1,5) combine (1,2)+(3,5): NP + VP -> S => top cell contains {S}.

Conclusion: Final cell (1,5) contains S, so the sentence is accepted by the grammar.

(iii) Final cell non-terminals: {S} — sentence grammatical.
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(iv) If we add VP → V: VP can be formed from a verb alone. This makes sentences like
"the dog chased" (NP + V) derivable: NP -> 'the dog', VP -> 'chased', S -> NP VP. It
increases language and can introduce ambiguity (multiple parses).
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Q1(b) Given grammar and sentence: "the teacher gave an apple"

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Apply CYK similarly. Tokens: 1=the,2=teacher,3=gave,4=an,5=apple.

Span 1:
(1,1): {Det}
(2,2): {N}
(3,3): {V}
(4,4): {Det}
(5,5): {N}

Span 2:
(1,2): Det + N -> NP => {NP}
(2,3): N + V -> none => {}
(3,4): V + Det -> none => {}
(4,5): Det + N -> NP => {NP}

Span 3:
(1,3): splits -> no NP V rule => {}
(2,4): {} => {}
(3,5): V + NP -> VP => {VP}

Span 4:
(1,4): {} => {}
(2,5): check (2,2)+(3,5): N + VP -> no rule => {}

Span 5:
(1,5): (1,2)+(3,5): NP + VP -> S => {S}

Conclusion: Final cell contains S => sentence accepted.

If VP → V is added, sentences like "the teacher gave" become acceptable (NP + V).
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Q1(c) (same grammar/sentence as Q1a)
Result: identical CYK table as Q1(a) leading to top cell {S}. Adding VP → V allows 'the dog chased' to become valid.

Q2. Semantic Disambiguation & Polysemy (a, b)

Q2(a) Lesk algorithm:
- Knowledge-based. For ambiguous word (bank), Lesk compares glosses (dictionary definitions) of each sense with the context words in the sentence.
- Compute overlap counts: tokens shared between gloss of sense and context (e.g., 'river', 'water' vs 'money', 'finance').
- Select sense with highest overlap. Example: 'crossing near the bank' — context 'crossing' and 'near' likely align more with riverbank senses in WordNet glosses.

BERT and HyperLex:
- BERT: contextual embeddings; represent the target word in its sentence embedding space; nearest-sense or classifier on embeddings identifies sense.

- HyperLex (graph-based): build co-occurrence graph of words in corpus, compute centrality/clusters for senses; senses manifest as dense subgraphs.

Comparison (interpretability, scalability, real-time):
- Lesk: interpretable, low compute, but needs gloss coverage and fails with sparse overlaps.
- HyperLex: unsupervised, scales with corpus size, interpretable via graph structure but needs good co-occurrence data.
- BERT: highly accurate, robust to noise, but heavy compute and less interpretable.

Q3. Anaphora Resolution and Coreference (a, b)

Q3(a) Example: "Meera called Riya after she finished her project. She thanked her for the help and invited her to the celebration."

(i) Noun phrases and pronouns:
- NPs: Meera, Riya, her (x2), She (x2)
- Pronouns needing resolution: 'she' (in both sentences), 'her' (twice)

(ii) Hobbs' algorithm (outline):
- Traverse parse tree left-to-right, propose antecedents by exploring noun phrases in sentence and previous sentences, checking agreement (gender/number) and binding.
- For the first 'she' (after she finished her project): likely antecedent is Meera (subject who called), but syntactic positions and recency must be considered. Depending on parse, Hobbs would find the nearest NP matching gender/number.

(iii) Centering algorithm:
- For each sentence compute Forward-looking centers (Cf) = ranked list of salient NPs, Backward-looking center (Cb) = preferred entity from previous sentence.
- Transitions (Continue, Retain, Shift) indicate coherence; pronouns typically refer to highest-ranked Cf.

(iv) Comparison:
- Hobbs is syntax-driven and often precise for local antecedents.
- Centering models discourse coherence better and predicts pronoun use patterns.

Q4. Multilingual Systems, Summarization, Sentiment, NER, QA

This section contains design pipelines. For brevity, key points:

Q4(a) Multilingual sentiment pipeline:
1. Language identification
2. Normalization (Unicode, transliteration)
3. Tokenization and morphological processing
4. Feature extraction (mBERT/XLM-R embeddings)
5. Classifier (fine-tuned transformer or lightweight classifier)
6. Post-processing and explainability

Challenges: code-mixing, low resources; use transliteration models, transfer learning, data augmentation.

Q4(b) Abstractive summarization:
- Preprocess, fine-tune encoder-decoder (T5/BART), use attention and beam search, apply factuality checks (QA-based or entailment filters), and post-edit for readability.

Q4(c) Speech-to-text summarization workflow:

- ASR (language-aware) → punctuation/diarization → language ID → segment → summarizer per language (multilingual model) → merge + coherence checks.

Q4(d–f) outlines:
- Headline generation: encoder–decoder, evaluate with ROUGE/BLEU and factuality checks.
- NER for Indian languages: use multilingual embeddings, character-level features, sequence models (BiLSTM-CRF or transformers).
- MT: multilingual transformer + domain adaptation; use terminology tables for domain-specific terms.
- QA: Retriever (dense + sparse) + Reader (T5/BART) + RAG for grounded generation; ensure reliability with oversight and source-ranking.

Q5. Structural Ambiguity and Shift–Reduce Parsing (a, b, c)

Q5(a) Sentence: "The man saw the hill with the telescope."
Grammar highlights: NP can be NP PP; VP can be VP PP → PP attachments lead to ambiguity.

Two interpretations:
1) PP attaches to VP: The man used a telescope to see the hill.
Parse: [S [NP The man] [VP saw [NP the hill] [PP with the telescope]]]

2) PP attaches to NP: The hill had the telescope.
Parse: [S [NP The man] [VP saw [NP [NP the hill] [PP with the telescope]]]]

Shift–Reduce sequences (outline):

Interpretation 1 (PP attaches to VP):
- SHIFT 'The' 'man' → reduce to NP (The man)
- SHIFT 'saw'
- SHIFT 'the' 'hill' → reduce to NP (the hill)
- SHIFT 'with' 'the' 'telescope' → reduce to NP -> PP
- Reduce: VP → V NP PP
- Reduce: S → NP VP

Interpretation 2 (PP attaches to NP):
- SHIFT 'The' 'man' → NP
- SHIFT 'saw'
- SHIFT 'the' 'hill' → NP
- SHIFT 'with' 'the' 'telescope' → PP
- Reduce NP → NP PP (attach PP to NP 'the hill')
- Reduce VP → V NP
- Reduce S → NP VP

Order of reductions changes structure and meaning. A shift–reduce parser that reduces earlier may prefer one interpretation; delaying a reduction may enable the other.

Q5(b) Coordination and attachment:
Sentence: "The boy and the girl saw the telescope with the lens."

Ambiguities:
- Coordination: 'The boy and the girl' could be parsed as NP -> Det N Conj NP or grouping differences.
- Attachment: 'with the lens' could attach to VP (saw with lens) or NP (telescope with lens).

Two shift–reduce parses:
1) Conjunction grouped (The boy) and (the girl) as coordinated subject; PP attaches to VP.
2) PP attaches to NP 'the telescope', giving different semantic roles.

Semantic cues (plausibility) can disambiguate: telescopes typically have lenses, so PP attaching to telescope is more plausible in world knowledge.

Q5(c) Example: 'I chased the dog with the stick.'
Two interpretations:
1) I used the stick to chase the dog (PP attaches to VP).
2) I chased the dog that had the stick (PP attaches to NP).

Draw parse trees as in Q5(a). Shift–Reduce sequences mirror the examples: attach PP early to NP for interpretation 2, or attach PP to VP for interpretation 1.

Q6. Naïve Bayes Word Sense Disambiguation (a, b)

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Q6(a) 'cell' — senses: room (room) vs biology (biological cell).
Given:
P(room)=0.6, P(biology)=0.4
Feature: 'microscope'
P(microscope|room)=0.05
P(microscope|biology)=0.3

Compute unnormalized posteriors:
score(room) = 0.6 * 0.05 = 0.03
score(biology) = 0.4 * 0.30 = 0.12
Normalize:
Z = 0.03 + 0.12 = 0.15
P(room|context) = 0.03/0.15 = 0.20
P(biology|context) = 0.12/0.15 = 0.80

Final answer: Predicted sense = BIOLOGY (0.80 posterior).
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Q6(b) 'bat' — senses: animal vs sport.
Given:
P(animal)=0.45, P(sport)=0.55
Feature: 'ball'
P(ball|animal)=0.1
P(ball|sport)=0.4

Compute unnormalized posteriors:
score(animal) = 0.45 * 0.10 = 0.045
score(sport) = 0.55 * 0.40 = 0.22
Normalize:
Z = 0.045 + 0.22 = 0.265
P(animal|context) = 0.045/0.265 ≈ 0.1698
P(sport|context) = 0.22/0.265 ≈ 0.8302

Final answer: Predicted sense = SPORT (≈ 83.0% posterior).
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Q7. WordNet Relations and Role (a, b)

Q7(a) Semantic relations:
- Hypernymy: generalization (dog → animal).

- Meronymy: part-of relation (wheel is part of car).
- Antonymy: opposites (hot ↔ cold).
- Troponymy: manner relations among verbs (to stride → to walk).

Polysemy/homonymy:
- WordNet stores senses as synsets; polysemous words have multiple synsets. Homonyms have distinct synsets with unrelated glosses.
- WSD uses synset distinctions to select sense based on context.

Q7(b) WordNet for inference:
- Entailment: 'snore' entails 'sleep' because WordNet links allow inference.
- Causation: 'buy' causes 'own' via relation chains and real-world knowledge.
Limitations: idioms, cultural context; BabelNet/WordNet++ expand coverage and multiword expressions.

Q8. Coreference & Discourse-level Resolution (a, b)

Q8(a) Examples: "Ravi helped Suresh to move his luggage before he left for the airport."
- NPs: Ravi, Suresh, his, he.
- Syntactic constraints: number, gender, reflexivity. Semantic constraints and world knowledge disambiguate (whose luggage?).
- Preferred resolution: 'he left' often refers to Ravi if Ravi is subject and last-mentioned salient, but context matters.

Q8(b) Priya/Neha example:
- 'She thanked her for the help.' Ambiguity: who thanked whom?
- Use recency, grammatical role, and pragmatic inference: if Priya met Neha after she submitted, and 'She thanked her' likely Priya thanking Neha (because Neha helped).
- Parallelism and repetition increase salience and bias resolution.

Q9. Naïve Bayes — Spam & Sentiment (a, b)

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Q9(a) Email classification (summary):
- Priors: P(Spam)=3/5=0.6, P(Ham)=2/5=0.4.
- Use multinomial Naïve Bayes with add-one smoothing.
- Test email: "The conference meeting was moved to 4pm."

Per-word likelihoods favor Ham (words like 'meeting', 'conference' appear in Ham
training).
After multiplying probabilities and normalizing:
P(Ham|email) ≈ 0.9167, P(Spam|email) ≈ 0.0833.

Decision: NOT SPAM (Ham).
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Q9(b) Sentiment classification (summary):
- Priors: P(Pos)=3/5=0.6, P(Neg)=2/5=0.4.
- Test review: "The plot was engaging, but the visuals were disappointing."
- Compute per-word likelihoods with smoothing and multiply.

Resulting normalized posteriors:
P(Positive) ≈ 0.4207, P(Negative) ≈ 0.5793.

Decision: NEGATIVE sentiment.
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Notes: Calculations for multinomial Naïve Bayes used add-one smoothing; CYK steps used unit-closure for NP $\rightarrow$ N. For full fraction-level arithmetic for Q9 or expanded CYK grid tables, request specific sections and I will append detailed numeric tables.